

SAM: Interface Design of a Mobile Application for Women in Risk Situations

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ABSTRACT

The following paper presents a design proposal of a mobile application aimed to support women in risk situations in the City of Huajuapán de León, Oaxaca, Mexico. Our proposal, Assistance System for Women (or SAM), is a design for women in a situation of risk. Based on an user-centered design approach, our proposal features functionalities like Help buttons, Simulate Telephone Calls and Follow-up, as well as Location in real time. Gender violence and violence against women is a factor of enormous concern in Mexican society nowadays. By creating this interface prototype, we intend to offer our users security to be able to carry out their daily activities without fear. We are trying to be part of positive social development, an essential element to achieve the maintenance of peace and security.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in collaborative and social computing.**

KEYWORDS

Risk situation, mobile application, Women's risk, Mexico, UN Goals, Peace, Justice and Strong Institutions.

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1 INTRODUCTION

In order to advance peace, justice, and strong institutions, the United Nations Sustainable Development Summit adopted 17 Sustainable Development Goals in 2015, building on the Millennium Development Goals. One of them, number 16 Peace, Justice and Strong Institutions seek to build inclusive institutions at all levels and peaceful societies with universal access to justice. This seeks to lessen the detrimental impacts of violence on health, development, well-being, and social inclusion while ensuring that individuals around the world do not live in fear of suffering violence, regardless of their background.

For our research, risk situations refer to “*actions that expose us to dangers and could cause damage to our health, whether physical or psychological, which can also affect the integrity of other people, and even cause the loss of life*” [12].

Latin America encompasses six of the first ten countries with the highest number of forced disappearances produced between 1980 and 2020, according to the Report of the Working Group on Enforced or Involuntary Disappearances of the United Nations Organization (UN) [11]. The Mexican Institute of Human Rights and Democracies A.C., highlights, in a diagnosis carried out on disappeared women that, “The general trend of cases of disappeared girls and women, from the year 2000 onwards, has been on the rise, reporting since 2011 more than 1,000 cases annually. As of 2020, more than 2,000 cases begin to be reported annually, and it is worrying that the highest peak would be in 2021, with 2,729 cases of missing and missing girls and women” [5]. According to the Mexican National Registry of Data on Missing and Unlocated Persons (RNPDO, 2022), it shows that between 2000 and 2022 alone, more than 19,000 women did not return to their homes [10]. Furthermore, the average age of these women was between 15 and 19 years old. The five Mexican states with the highest number of missing women are: State of Mexico, Tamaulipas, Jalisco, Mexico City and Veracruz. During 2022, the incidence of complaints in the state of Oaxaca for disappearances of girls, adolescents and adult women is 159 cases,

registering an upward trend. Of them, so far 84 have been found alive, one more lifeless and 74 have not been located [8].

2 BACKGROUND

We started by researching which apps already exist in the area and what they offer to the community, aimed to help to solve this serious problem.

Mobile phones are currently an essential part of daily life, it is perhaps the first object we pack when leaving our home. At present, it is impossible to think of ourselves without the presence of these telephones, they condition our day to day, because through them we communicate, schedule appointments, remember to take medicines or, simply, they serve as entertainment centers, among other things. [14].

Currently, there are applications able to launch panic buttons and SOS alerts. These applications track the user's location in real time, and send help information to a certain number of contacts when an alert is activated. Some of these applications are described below [15]:

- **SOSMex**: It is an app in which you can register up to three trusted contacts and share your real location at all times. Also, if you are at risk, it works as a panic button.
- **Sister**: The objective of Sister is to respond immediately in case of emergency, it also has geo-location feature to share the location in real time, as well as the arrival on time to any place; otherwise, an alert will be activated that will be sent to your trusted contacts.
- **SOSFem**[13]: An application that helps women in dangerous situations, consists of a panic button which is activated in three ways: Motion sensor, (pressing the lock button five times directly from the application). By activating the button, your emergency contacts receive an SMS with a link to your location.

In recent years, an increase in the interest for creating projects that help women in risky situations has been observed. One of these projects with this issue is analyzed below to identify trends and best practices implementations for these projects.

- **Anshimi**, a mobile application that hosts security services for women in Seoul, South Korea, provides monitoring and safe return exploration services. The monitoring service uses GPS data from mobile phones and a security camera tracking to provide rescue support to women in case of an emergency[16].

Similarly, there are gadgets that can help to prevent and reduce violence against women. Some examples we have found are:

- In 2022, Mexico implemented a **free SOS bracelet** program to combat violence against women. This project includes a portable device that features a panic button, which allows victims of gender-based violence to send an alert signal to selected contacts and the nearest police station. This bracelet can also be used as a key chain and is provided free to women who have protection measures issued by specialized prosecution offices for family and gender violence[9].

- During the 9th International Conference on Networking, Systems, and Security (NSysS) in 2022, a **wireless Internet of Things (IoT) device** was presented to provide security for women in risky scenarios. The devices communicate using IoT and monitor vital parameters such as pulse and temperature and are automatically activated if a dangerous situation occurs. Its functions include sending alerts via text using the GSM cellular network, real-time location coordinates through GPS, and generating an alarm wirelessly to transfer sensor data to an open-source cloud platform that allows tracking of current data[7].
- In the thesis "*Design of a Smart Bracelet*" which proposes implementing a neural network in a device to detect changes in vital signs that may indicate stress or danger, a Bluetooth module will send the data to the smartphone application. The application will send a distress alert and the real-time location of the person in danger to registered numbers[1].
- **SafeBand** is a security device to combat physical harassment against women in Bangladesh. The device consists of a portable wristband or necklace that has a button and a light. When the button is pressed, the user's location is identified through the global positioning system (GPS), and a message is sent to the nearest police station and previously saved contacts (family members' numbers). The system also includes two mobile applications, one for the victim and one for the police. The article describes an empirical study conducted to evaluate the effectiveness, efficiency, usefulness, and acceptance of the system[6].

In summary, there are various technological advances that have been specifically designed to help women in situations of violence, such as mobile safety applications, personal alert devices, and surveillance cameras. This indicates that it is a problem that is being addressed from different perspectives that have been developed in different contexts.

The mentioned applications are of great help to users, as they allow them to send distress messages and share their real-time location with their trusted contacts. However, when user testing was conducted with some of these applications, specifically with SOSFem, we found that the time required to perform tasks such as registering in the application took too long. In addition, the interfaces did not provide enough information on how to use them (usability issues). There are also personal security devices that help women feel safer while they are away from home. These devices include personal alarms, tracking and monitoring devices, and self-defense devices.

3 METHODOLOGY

User Centered Design (UCD), defined by the UX Professionals Association (UXPA), as a design approach whose process is directed by information about the users who are going to use the product, is our methodology of choice [4].

UCD seeks to ensure the achievement of a product with the appropriate functionality for specific users. The objective of this philosophy is to offer answers to questions such as: Who will use this system? What will they do with it? What information will you need to achieve your objectives?.

The main difference of UCD compared to other approaches is that its process is not sequential or linear, but presents cycles (Figure 1) in which the design is iteratively tested and optimized until the required level of quality is reached.

We started the first phase of User-Centered Design with the application of a survey to learn about our users. We created wireframes and shared them in a Focus Group to obtain feedback, and we returned to the design phase and improved our screens with a prototype. Based on the information obtained from the Focus Group, we finally evaluated our prototype, the phases developed are detailed below.

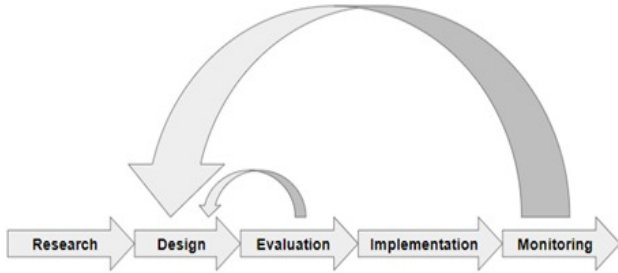


Figure 1: Diagram of the User-Centered Design process. Adapted from User Experience: Principles and Methods (p. 16), by Hassan, M, 2014.

3.1 Research

Our research stage began with the application of a survey to 166 women, these women live in the City of Huajuapán de León (one of the cities with the highest index of violence against women in the Oaxaca Mixtec region [17]). Their age range is mostly between 15 and 18 years of age. Just over 91.4% are students.

Some of the applied questions are listed below:

- Have you suffered risk situations?
- Where have you felt insecure?
- What do you do when you feel in danger?
- Have you had any help tools or systems?
- Have you used any help application for these risk situations?
- What has been your experience using these applications?
- If you could design the perfect helpdesk app, what features would it have?

Analyzing the responses, we obtained the following insights:

From our respondents, 105 of them affirmed having suffered some type of risk situation. Regarding the places where they have felt insecure, responses that mention public transport and the street prevailed, followed by school. On these occasions, 82 of them responded that they had been alone, only 18 stated that they were accompanied, and the rest said that they had been alone and accompanied.

Most of them coincide in looking for public places to feel safe, while some of them mention that they call a relative or send their location to their acquaintances, the latter indicating that a considerable part of them use the cell phone as a help tool.

Only 17 of the respondents who have suffered risk situations mentioned using a help application, with which they have not had a

good user experience, as some mentioned that they did not know how to use the application well or because it takes a long time to respond.

Finally, we asked the participants to come up with the perfect application for this type of situation, and what features they would like to have. Of the 105 responses we got, most of them responded that they would like to have the following features:

- Easy to use
- Geo-location
- SOS messages
- Media recording
- Emergency calls to relatives

3.2 Design

Once the analysis of the questionnaire was finished and having already a broader context about the users, and the characteristics that the participants would like to have, the development of wireframes began.

We took into account all the characteristics that the participants had answered. With this, we defined the main actions and tasks that the user should do, taking care of the distribution of the elements. The wireframes that were designed with respect to the characteristics said by the users can be seen in Figure 2. This figure shows the wireframes that were obtained based on the characteristics desired by the respondents.

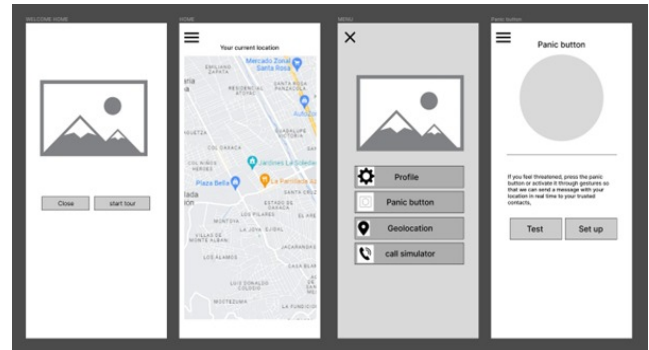


Figure 2: Wireframes of the main SAM screens. Source: our own.

To obtain opinions about the wireframes, the technique of Focus Groups (Figure 3) was used. From Escobar and Bonilla-Jiménez (2017), "focus groups are a data collection technique through a semi-structured group interview, which revolves around a theme proposed by the researcher" (p.2). The main purpose of the Focus Group is to bring out attitudes, feelings, beliefs, experiences and reactions in the participants, something that would not be easy to obtain with other methods [3]. Using this technique, a series of questions about the wireframes were applied to a group of 5 women, but not before using techniques to break the ice. Among these questions are the following:

- What would you add to this screen?
- Would you like to add anything else?
- What do you think of this screen?
- How do you think this activity is carried out?



Figure 3: Focus Group with users. Source: our own.

We discovered what users thought of actions such as the Panic button or the Call Simulator should be executed using a shortcut or a gesture within the cell phone; looking to do this in a hidden, and fast way.

With this information, new fields were added in the wireframes of the different tasks to be able to relate a gesture with the execution of a feature, for example, relate the gesture of "pressing the power button for 5 seconds" to send a help alert.

The design of the screens was carried out in the Figma (desktop version) and thanks to its prototyping tools, interaction flows were added so that when simulated in the mobile version of the program, they would look as real as possible (Figure 4). This figure shows some of the main screens of our design proposal.

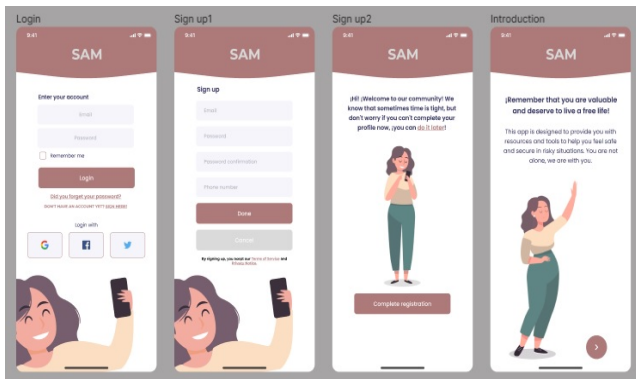


Figure 4: Screenshots of the proposed interface designs for the SAM mobile application. Own source.

3.3 Evaluation

Finally, usability tests were conducted with our four participants who meet the characteristics of our target user, that is, young female students between 15 and 19 years old. The recipient user (trusted contacts) was excluded as they do not directly interact with the application.

The evaluation phase for the prototype focused on assessing usability, considering the following tasks:

- (1) Register in the application
- (2) Configure the data of your Help button.
- (3) Set up your fake call.
- (4) Edit your profile.

It is important to mention that any usability test should be conducted in such a way that the integrity of the users is not compromised, with the protection of the rights and well-being of the participants as a priority. Recreating a real situation of violence could constitute impediments, and implies exposing the participants to potentially traumatic experiences, which could be harmful to their physical and psychological well-being.

With the all above, it was possible to recruit users to perform usability tests in the usability lab. It should be noted that these tasks did not require scenarios where a real violence situation was recreated. Despite this, the real context was considered as far as the study and design covered.

We highlight that two of the participants already had experience using similar applications, while the rest had no prior experience of having used any application. All participants had to have time to carry out the test in the UsaLab Laboratorio de Usabilidad in our university. The tests were guided by two facilitators (Figure 5) who told the participants the scenarios and the tasks they had to perform.

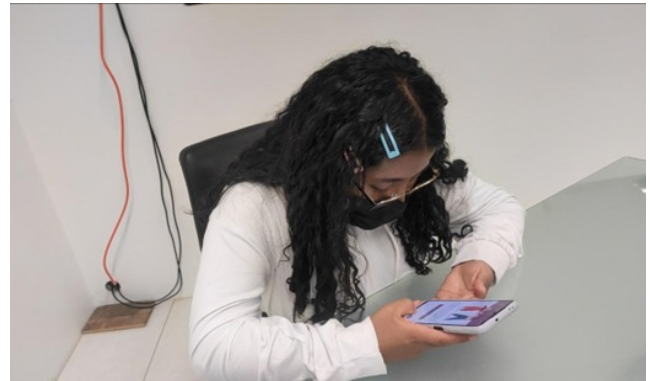


Figure 5: Usability test of our design proposal with a user.

The most widely used or recognized definition of usability is the one set forth in the ISO 9241-11 standard, in which usability is described as the degree to which a product can be used by specific users to achieve specific objectives effectively, efficiency and satisfaction, in a context of specific use [2].

The ISO 9241-11 standard defines satisfaction as: "absence of discomfort and existence of positive attitudes towards the use of the product". On the other hand, according to ISO 9126-1, satisfaction

is defined as the software's ability to meet user expectations in a given context of use.

With the help of a screen recorder, we recorded the user's interaction with the prototype from start to finish during the test. At the end of each test, a user satisfaction survey was conducted.

What we discovered from the tests carried out was that:

- Most of the participants successfully completed the asked tasks.
- Users point out that they would like to have buttons to choose which functionalities of the application they want to activate and thus perform the corresponding configuration.
- Regarding the information requested for functionalities, such as the help button or call simulator, they mentioned that filling in the fields was not cumbersome for them.
- Users with experience in similar tools noted the ease of use of our interfaces, telling us that the prototype they were presented with was visually appealing compared to what they had already used.
- They highlighted the use of a character to guide them through their first interaction with the system. However, 1 of the 4 users claimed that the color palette used in the design was not very striking.

4 RESULTS

Users of the usability test reported a feeling of a sense of promise of security when using this application. Also, many of them mentioned that the navigation and layout of the options was intuitive and easy to follow, giving them confidence in their ability to handle the application successfully. Additionally, several users noted that the presence of security measures such as user authentication and data encryption gave them a sense of peace when using the application for sensitive transactions. Overall, users felt positively impressed by the sense of security they experienced when using this application.

Regarding the proposed SAM design, it has been proven to be effective in improving the user experience for women. It has focused on usability and accessibility to ensure a user-friendly interface, allowing women to use the system efficiently and without difficulty. Additionally, it has been proven that the system is easier to handle than existing applications on the market, making it more attractive for use.

However, some limitations have been identified in terms of the need for users to add selectable items. This can be considered as an issue to improve for future versions of the system.

Lastly, the survey conducted at the end of the tests showed that users had a positive opinion about the system, with most describing a pleasant experience. Users showed enthusiasm for the incorporation of a character who guides them during their first time on the system screens, which helps to improve the user experience. In general, it can be concluded that the system has been designed effectively and is user-friendly for users.

5 CONCLUSION

In conclusion, the results of the usability testing and surveys conducted have shown that the proposed design of the SAM is effective in improving the user experience for women, as it has focused on usability and accessibility to ensure a user-friendly interface. Additionally, it has been proven that the system is easier to handle than existing applications on the market, making it more attractive for use. However, some limitations have been identified in terms of the need for users to add selectable items, which can be considered as a point to improve for future versions of the system.

The implementation of usability design methodologies in the development of systems, such as the SAM, can have a significant impact on reducing costs and development, maintenance, redesign, and support times, while also improving credibility by increasing user satisfaction and trust, improving performance by reducing the number of user errors, and increasing ease of use and learning. Additionally, by improving the user experience for women, we are also contributing to the fulfillment of Goal 16 of the United Nations, which aims to promote gender equality and empower women and girls. In general, it can be concluded that the proposed design of the SAM is an effective and promising solution for improving the user experience for women and contributing to the achievement of gender equality.

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